

Materials

Task Query Q

Solve this math problem:
An electric motor has a label ...

proposed by GPT-4

Collaborative Group Pool $\mathcal{M}$

M_1 : Code Group

Roles:{Analyst, Coder}
Topology:{Chain}



M_2 : Reason Group

Roles:{Expert, Solver, Critic}
Topology:{Chain}



M_3 : Debugging Group

Roles:{Coder, Checker, Plan}
Topology:{Star}

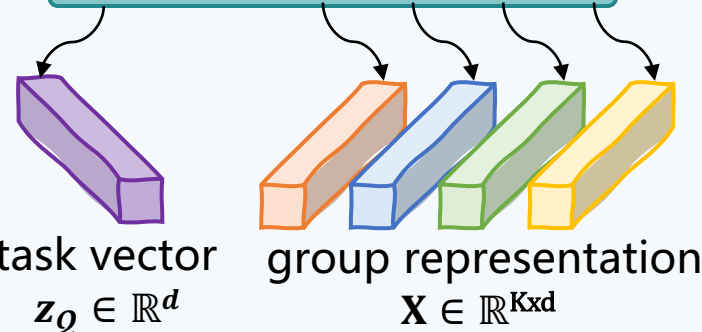


M_4 : Eval Group

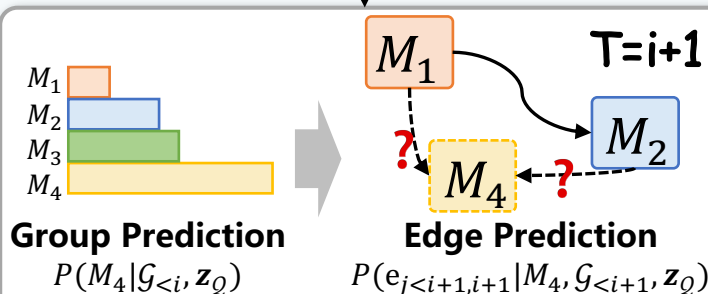
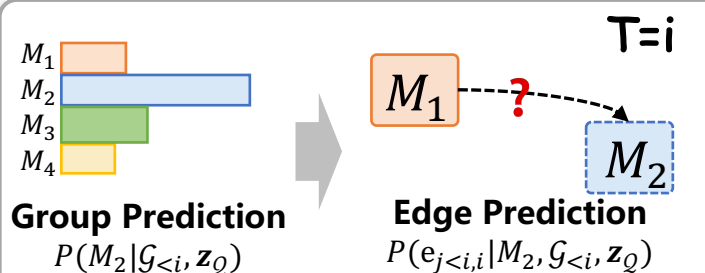
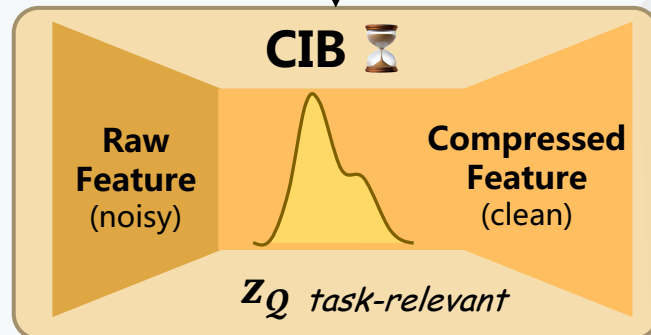
Roles:{Analyst, Math, Critic}
Topology:{Full Connected}



Sentence Encoder

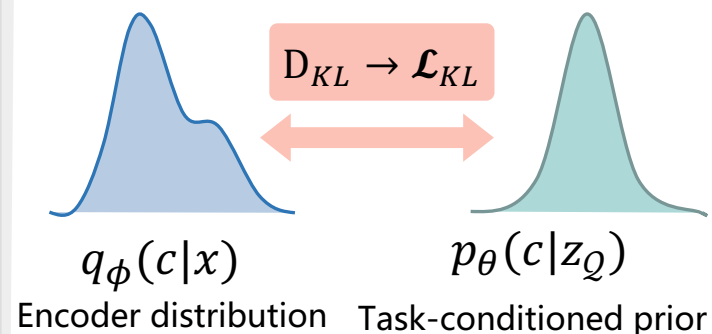


Design



Optimize

Zoomed-in of CIB



*Standard IB compresses blindly;
CIB compress conditioned on the task
only removes what's irrelevant to Q*

Group Prediction $\mathcal{L}_{group}$

Edge Prediction $\mathcal{L}_{edge}$

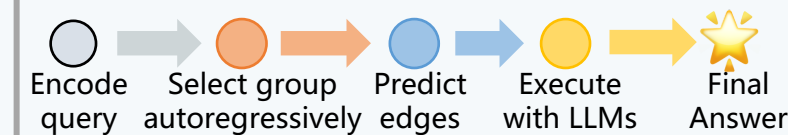
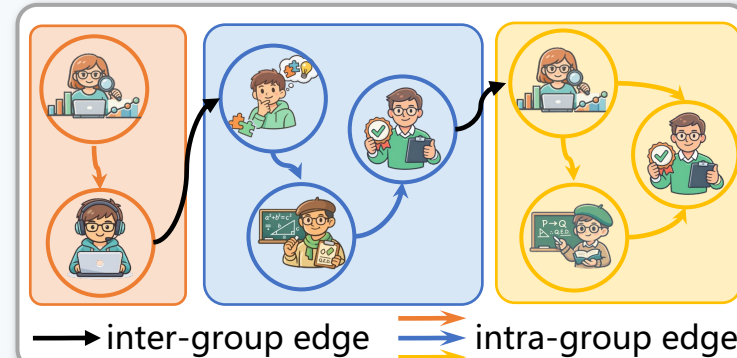
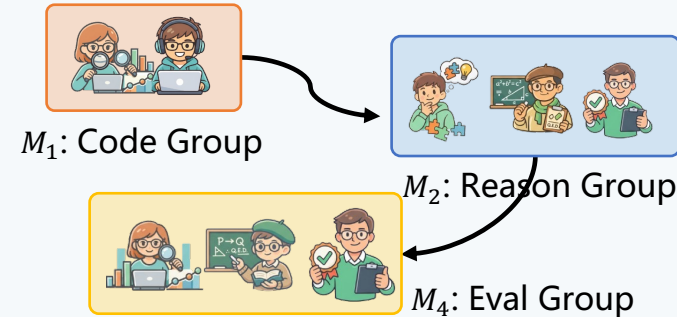
$$\mathcal{L} = \mathcal{L}_{group} + \mathcal{L}_{edge} + \beta_g \mathcal{L}_{KL}^{group} + \beta_e \mathcal{L}_{KL}^{edge}$$

Predictive Term Compression Term

β warm-up during training

Execution

Generated Topology $\mathcal{G} = (\mathcal{M}, \mathcal{E})$



Output Solution

Based on the reasoning and eval,
the solution to the math problem
is [Answer Details]

Efficient: fewer tokens than baselines.